

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	26 July 2026
Team ID	
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

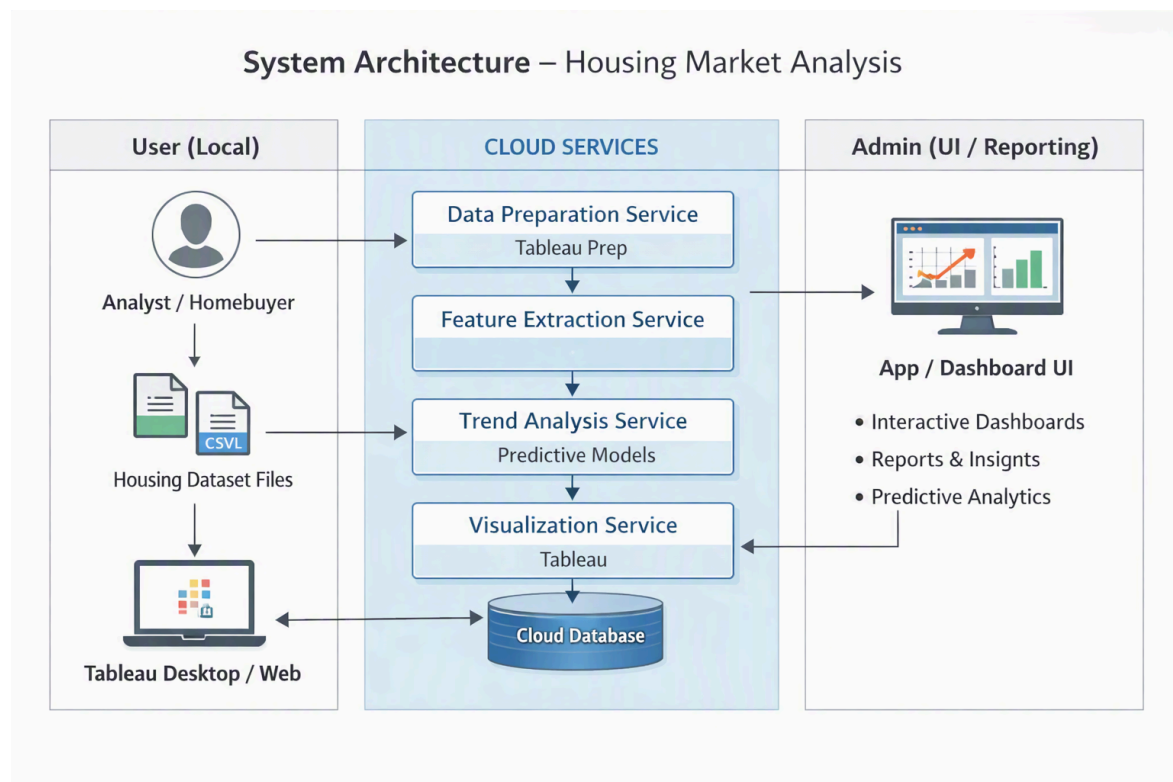


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Desktop Dashboard	HTML, CSS, JavaScript, React JS, Tableau Interface
2.	Application Logic-1	Logic for importing, cleaning, and preparing housing datasets.	Python (Scripts / Tableau Prep Flows)
3.	Application Logic-2	Logic for analyzing sale prices and property features.	Tableau Calculated Fields / Python Analytics Scripts
4.	Application Logic-3	Logic for predictive modeling and trend forecasting.	Machine Learning Models (Python / Scikit-Learn / TensorFlow)
5.	Database	Stores raw and processed housing data.	MySQL, NoSQL, etc.
6.	Cloud Database	Database service hosted on cloud for scalability.	IBM DB2, IBM Cloudant etc.
7.	File Storage	Stores uploaded datasets and exported reports.	IBM Block Storage or Other Storage Service or Local Filesystem
8.	External API-1	Fetches real-time housing or location data.	IBM Weather API, etc.
9.	External API-2	Retrieves demographic or economic indicators for analysis.	Aadhar API, etc.
10	Machine Learning Model	Predicts housing prices based on features and trends.	Object Recognition Model, etc.
11	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Python (Pandas, NumPy, Matplotlib), React JS, Tableau Public
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	SHA-256 Encryption, IAM Controls, SSL Certificates
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	3-Tier Architecture / Microservices on IBM Cloud
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Load Balancers, IBM Cloud Auto-Scaling
5.	Performance	Design consideration for the performance of the application	Caching Mechanisms, CDN Integration, Tableau Performance Optimizer

S.No	Characteristics	Description	Technology
		(number of requests per sec, use of Cache, use of CDN's) etc.	